

KAMAL GUPTA

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Research Statement: Extending Large Multimodal Models into the physical dimension. My goal is to enable embodied AI agents to reason, plan, and act in the real world with the same fluidity as digital agents through long-range video understanding and robust post-training.

WORK EXPERIENCE

Tesla AI (Optimus), Staff Research Scientist

Palo Alto, May 2023 –

- **Learning from humans:** Training large-scale multimodal models with human-centric physical tasks, with Imitation Learning and RL from human demonstrations to achieve zero-shot transfer to the Optimus
- **Long-Range Video Understanding:** Developing architectures and training recipes to improve the temporal context window of vision-language models, enabling robots to reason over long-horizon tasks (minutes vs. seconds)
- **Physical Evals:** Developed rigorous evaluation suites (sim and real) to quantify model reasoning, causal understanding, and safety in unstructured environments for humanoid robots.

Google Research, Research Intern

Mountain View, May 2022 – Nov. 2022

- **Spatial Alignment for Foundation Models:** Developed ASIC (Aligning Sparse in-the-wild Image Collections), a framework to bring order to unstructured visual data by establishing spatial and temporal consistency across disparate views (accepted to ICCV'23)

NVIDIA AI, Research Intern

Santa Clara, May 2021 – Nov. 2021

- **Discrete Latent Representations:** Engineered two-stage generative models using VQ-VAE style architecture to synthesize complex textured meshes, focusing on preserving physical and geometric constraints.

Amazon AWS, Research Intern

Pasadena, May 2019 – Aug. 2019

- **Foundational Generative Modeling:** Early research in adapting decoder-only autoregressive Transformer architectures (GPT-style) for structured visual domains (3D, documents, wireframes), demonstrating that "digital-first" architectures are highly effective for spatial reasoning (paper accepted in ICCV'21)

NetraDyne, Staff Research Engineer

Bengaluru, Mar. 2017 – Aug. 2018

Netradyne provides ADAS devices for commercial vehicles. I led a team of 4 to ship

- **Temporal Video Reasoning:** on-device models to interpret long-range temporal cues (distraction, drowsiness, and complex driving maneuvers) in varied environments, managing the "long-tail" of real-world physical data.
- **Large-Scale Behavioral Alignment:** a real-time safety pipeline to align driver behavior with human safety standards, processing millions of driving miles to refine edge-case detection.

Poolka AI, Cofounder, CTO

Bengaluru, Apr. 2016 – Feb. 2017

- **Full-stack Multimodal Model Deployment:** Designed and shipped Fairi, an early chatbot agent that integrated visual understanding (pose, segmentation) with language models to provide fashion recommendations.

American Express AI Labs, Research Engineer

Bengaluru, July 2013 – Mar. 2016

- Large scale recommendation systems, Geometric deep learning on financial data.
- **Platinum Genius Medal** - Systems and methods for customized real time data delivery Dec. 2015
- **Trainer of the Quarter** - Hadoop course for >200 AmEx employees in NYC, Gurgaon, Bangalore Dec. 2014

Robotics Institute, Carnegie Mellon University, June 2011 – June 2012 Research Intern, Pittsburgh, PA

- Developed an approach to predict vineyard yields automatically and non-destructively with cameras ([web](#))
- Estimated global pose of MAV using stereo visual odometry fused with infrequent GPS measurements ([web](#))

EDUCATION

University of Maryland, College Park

Aug. 2018 - May 2023

Ph.D. in Computer Science (4.0/4.0), Kulkarni fellow, Dean's fellow, Outstanding GRA Award

Thesis: Learning and Composing Primitives for the Visual World

Indian Institute of Technology Delhi

Aug. 2007 - Dec. 2012

Bachelors + Masters in Electrical Engineering (8.6/10.0)

Thesis: Pose Estimation for a Micro-Aerial Vehicle in GPS-denied environments

SELECTED PUBLICATIONS AND PATENTS

Visit Google Scholar [tC3td8cAAAAJ](https://scholar.google.com/citations?user=tC3td8cAAAAJ) for the complete list (* denotes equal contribution)

- Measuring Style Similarity in Diffusion Models *ECCV* 2024 [[web](#), [code](#)]
G. Somepalli, A. Gupta, **K. Gupta**, S. Palta, M. Goldblum, J. Geiping, A. Shrivastava, T. Goldstein
- LiFT: A Surprisingly Simple Lightweight Feature Transform for Dense ViT Descriptors *ECCV* 2024 [[web](#), [code](#)]
S. Suri*, M. Walmer*, **K. Gupta**, A. Shrivastava
- SHACIRA: Scalable HAsH-grid Compression for Implicit Neural Representations *ICCV* 2023 [[web](#), [code](#)]
S. Girish, A. Shrivastava, **K. Gupta**
- Chop & Learn: Recognizing and Generating Object-State Compositions *ICCV* 2023 [[web](#), [code](#)]
N. Saini, H. Wang, A. Swaminathan, V. Jayasundara, B. He, **K. Gupta**, A. Shrivastava
- ASIC: Aligning Sparse in-the-wild Image Collections *ICCV* 2023 [[web](#), [code](#)]
K. Gupta, V. Jampani, C. Esteves, A. Shrivastava, A. Makadia, N. Snavely, A. Kar
- Teaching Matters: Investigating the Role of Supervision in Vision Transformers *CVPR* 2023 [[web](#), [code](#)]
M. Walmer*, S. Suri*, **K. Gupta**, A. Shrivastava
- LilNetX: Lightweight Networks with EXtreme Compression & Structured Sparsification *ICLR* 2023 [[web](#), [code](#)]
S. Girish, **K. Gupta**, S. Singh, A. Shrivastava
- Neural Space-Filling Curves *ECCV* 2022 [[web](#), [code](#)]
H. Wang, **K. Gupta**, L. Davis, A. Shrivastava
- PatchGame: Learning to Signal Mid-level Patches in Referential Games *NeurIPS* 2021 [[web](#), [code](#)]
K. Gupta, G. Somepalli, Anubhav, V. Jayasundara, M. Zwicker, A. Shrivastava
- LayoutTransformer: Layout Generation and Completion with Self-attention *ICCV* 2021 [[web](#), [code](#)]
K. Gupta, J. Lazarow, A. Achille, L. Davis, V. Mahadevan, A. Shrivastava
- Improved Modeling of 3D Shapes with Multi-view Depth Maps *3DV* 2020 [[web](#), [code](#)]
K. Gupta*, S. Reddy*, K. Shah*, A. Shrivastava, M. Zwicker
- PatchVAE: Learning Local Latent Codes for Recognition *CVPR* 2020 [[web](#), [code](#)]
K. Gupta, S. Singh, A. Shrivastava
- Applying Multi-Dimensional Variables to Determine Fraud [USPTO 16/426826](#), 2019
K. Gupta, V. Jain
- Systems and methods for Updating Fraud Detection Variable [USPTO 15/258880](#), 2018
K. Gupta, V. Jain
- Systems and methods for customized real time data delivery [USPTO 14/961614](#), 2015
S. Sanyal, S. Purkayastha, T. Choudhuri, A. Choudhary, V. Grover, M. Naeem, P. Mehta, **K. Gupta**, A. Agarwal

VOLUNTEER INITIATIVES

- **Graduate Student Mentor:** [CVPR Academy](#) for first-time CVPR attendees 2022
- **Co-organizer:** [SIGGRAPH RCDC](#) mentorship program for Graphics graduate school applicants 2021
- **Instructor:** [AI4ALL](#) summer program for high school students 2020
- **Reviewer:** Computer Graphics - SIGGRAPH 2022, SIGGRAPH Asia 2023. Computer Vision - ECCV 2022. CVPR 2023, 2022*, 2021. ICCV 2021. Machine Learning - NeurIPS 2022, 2023. ICLR 2022. (* denotes Outstanding Reviewer Award)

MISCELLANEOUS

- Kulkarni Research Fellowship ([web](#)) 2020
- Dean's fellowship for outstanding academic achievement 2018-20
- **Coordinator:** [Robotics Club](#) IIT Delhi 2008-10
- Master's Research Scholarship by Ministry of Human Resources and Development, Govt. of India May 2011
- CBSE Merit Scholarship for All India Rank 38 (600,000 appeared) in AIEEE, now called JEE-Main 2007
- National top 1% (40,000 appeared) in Physics Olympiad and Chemistry Olympiad 2006
- National Talent Search Exam Scholarship by Govt. of India 2005